

A NEW PERSPECTIVE ON THE INNER EAR

ELH Studio

Operation Manual

Visualization and quantification of endolymphatic hydrops on MRI

PPI − PEI = HYDROPS

Startup and basic workflow

ELH Studio Operation Manual

ELH Studio 1.3.1 (formerly HYDROPS Studio) | Updated November 2, 2026
Scope: research image processing using PPI, PEI and MR cisternography (MRC)

First steps

Extract the entire distribution ZIP and open ELH_Studio.html in Google Chrome. If Chrome is the default browser, double-click the file; otherwise right-click the HTML file and choose Chrome (on Windows, Start_ELH.bat in the same folder also opens it in Chrome). No Python or server is needed.

Welcome screen and display settings

The welcome screen shows an inner-ear motif; “Get started” opens the work screen. To hide it, check “Do not show on startup” or turn it off in “Settings” (top right), which can show it again anytime. Settings also cover ROI transparency, unsaved-change warnings and hiding identifying information (page 9), and are kept in the browser.

Basic workflow

1. Click “Select DICOM folder” and choose the folder of the whole study.
2. In the header list, choose the study and the PPI, PEI and MRC series (page 2).
3. Click “Create images from selected series” and wait for processing to finish.
4. Check misregistration and ROIs in the three columns; correct them if needed (pages 3–6). LUMEN: page 15 on.
5. Click “Save results” to download a ZIP, and check where it was saved (page 7).

Contents of the distribution folder

File	Purpose
Start_ELH.bat	Windows launcher that opens the HTML in Chrome; keep it in the HTML's folder
ELH_Studio.html	The app. Open it in Google Chrome
操作マニュアル.pdf	This manual (Japanese)
Operation_Manual_EN.pdf	This manual (English; same chapter order and page numbers as the Japanese)
THIRD_PARTY_NOTICES.txt	Licenses of the libraries used
配布版情報.txt / SHA256SUMS.txt	Version, scope of verification and file checksums

Supported browser and verification status

Google Chrome only; Microsoft Edge, Safari, Firefox and other browsers are not supported. In them, the notice “ELH Studio works only in Google Chrome” appears before the work screen; “Continue anyway (not supported)” proceeds, but operation is not guaranteed. The choice is not saved, so the notice appears at every opening.
BAT launch and direct opening of the HTML (file://) on an actual Windows PC are unverified. Screens were checked in Chrome on macOS. With the first case, check loading, display, saving and reopening.

Selecting series from DICOM headers

1. Load the folder of the whole study

Click “Select DICOM folder” and choose a folder that contains several series. If the browser asks to read the files in the folder, check the selected folder and allow it. Processing takes place in the browser.

2. Choose the study

Check the target in the study selector. Labels such as Study 01 are serial numbers assigned in this session, not patient IDs. Dates and series names are hidden by default; if you need them for matching, turn the hiding off in “Settings” (top right) and reopen the series selection screen.

Reading the header list

Item	What to check
Series number and name	Series Number / Series Description. Match them with the sequence name and purpose
TI / TR / TE	Inversion time, repetition time and echo time (ms)
Slices and matrix	Number of slices and image size (columns × rows)
Pixel spacing	In-plane pixel spacing (mm); not the slice spacing
Image Type	Attributes such as original or derived image. Do not decide by the name alone

3. Assign PPI, PEI and MRC

Choose a series in each selector. Candidates are suggested from series names (PPI / PEI / MRC / CISTERN) and values such as TI 2250 ms for PPI and TI 2050 ms for PEI, based on the imaging parameters used in development; they are not a rule for every protocol. With several candidates, none is selected automatically.

Check before creating images

Make sure PPI and PEI are not swapped and no series is assigned twice. Assign MR cisternography to MRC. Match series names with imaging parameters, and do not choose an existing HYDROPS image or another subtraction image as PPI or PEI.

4. Start display and calculation

With the automatic-processing box checked, “Create images from selected series” runs automatic localization, segmentation and measurement when MRC is present. Unchecked, it only creates the images; you can run “Detect & measure” later. The reference template is an average of 6 cases of the public dataset VS-SEG and can miss on other scanners and protocols; check the seed points and correct them manually if they are off (page 14).

When no series is found

Even without candidates, you can choose series manually from the header list. If processing stops because of the format or a coordinate mismatch, see page 8. Matching headers do not mean that motion between acquisitions is corrected.

Reviewing the three images side by side

Screen layout (sketch)



Synchronized display controls

Action	Effect
Wheel / slider below	Moves the slice of all three images together
Drag on an image	Pans all three images together (Inspect / pan mode)
+ / −, Ctrl + wheel	Zooms in and out in sync
Fit view	Returns to the full view; the slice position is kept
Right inner ear / Left inner ear	Zooms to the side with seed points and moves the slice
Selector above each image	Switches among PPI / PEI / HYDROPS / MRC / HYDROPS-Mi2
ROI overlay / Seed points	Shows or hides the ROI colors / the seed markers
Contrast	Changes the display only; source signal values and measurements are kept

Comparing PPI, PEI and HYDROPS

Check that the position and shape of the inner ear correspond between PPI and PEI, and compare them with the distribution of negative signal on HYDROPS. Switching to MRC shows the position and outline of the candidate total fluid space. Turning the ROI overlay off once makes the source image easier to see. Copying images and volume ratios: page 10.

Colors and orientation

Blue: cochlea candidate; green: vestibule candidate; red: negative HYDROPS signal within the ROI (candidate endolymph). Red is drawn over the region color. Check the R / L / A / P labels on the screen. The display follows the acquisition planes and may be slightly oblique.

While editing a contour

While you draw a contour, clicks add vertices. To drag the image, set “Tool” back to “Inspect / pan”. Changing the slice discards an unapplied contour.

Checking measurements and ROIs

Formulas

HYDROPS = PPI − PEI
HYDROPS-Mi2 = (PPI − PEI) × MRC
Volume ratio (%) = negative-signal voxels in the ROI ÷ all ROI voxels × 100

Checking each cochlea and vestibule

Four cards show the volume ratio over the whole 3D region, negative-signal volume / total volume (mm³), the area ratio on the current slice, the number of slices containing the ROI, and the values at z−1 / z+1. Clicking a card moves to that region's seed point. “—” for a missing ROI does not mean 0%.

Telling the results apart

Display	Meaning
The large %	Negative-signal volume ratio over the whole 3D candidate ROI
Current slice %	Area ratio within the slice now displayed
z−1 / z+1	Sensitivity analysis with a one-slice shift; not a confidence interval
Flags such as a high negative-signal ratio	Prompts to review contamination, boundaries or images; not a diagnostic judgment

Order of visual review

1. Check misregistration between PPI and PEI. Matching coordinates alone do not rule out motion.
2. On MRC, check on every slice that the blue and green ROIs stay within the target structures.
3. Check for contamination by the internal auditory canal (IAC), CSF or semicircular canals, and for missing parts.
4. Check that the red negative signal lies in plausible places within the candidate ROI; correct it if needed.
5. After the review, check “I reviewed the ROIs and image alignment visually” and save.

Scope of the judgment

The check box records a visual review; it does not confirm a diagnosis. Automatic processing and ROI edits clear the review status. No vestibular cross-sectional-area grade threshold is applied to the 3D volume ratio, and no cochlear morphological grade is assigned automatically.

Methodological notes

Negative signal means below 0; 0 is not included. Multiplying by a positive MRC does not change the sign, so for the same ROI the negative-signal ratios of HYDROPS and Mi2 are essentially the same. The subtraction image PPI − 0.04×MRC (HYDROPS2 of input type B, page 26) is calculated differently.

Editing ROIs

Starting an edit

Click “Edit ROIs” and choose right cochlea, right vestibule, left cochlea or left vestibule in “Region to edit”. An ROI is the region that is measured. To keep the state before the change, first save a ZIP with “Save results”.

A. Resegmenting from seed points

1. Under “Tool”, choose “Place seed → resegment”. The left image switches to MRC.
2. Adjust the slice and zoom, and click on the MRC fluid signal of the selected region.
3. After automatic segmentation the four regions already have seed points, so the ROI of each region is resegmented.
4. If automatic localization failed, place seed points for all four regions; segmentation runs once all four are placed.

Caution on resegmentation

Changing a seed point resegments all four regions, so earlier contour edits may be replaced. Correct seed points first, then refine contours. Running “Detect & measure” again also replaces the existing ROIs and the contour undo history.

B. Adding or removing on the current slice with a contour

1. Choose the region and the slice, then choose “Add ROI with contour” or “Erase ROI with contour”.
2. Click around the target area to enclose it with at least 3 vertices.
3. Press “Apply contour” or Enter to apply. Before that, “Cancel contour” or Esc discards it.
4. Check the adjacent slices too, and edit each slice that needs it. Contours are not interpolated to other slices.

Undoing after applying

“Undo last ROI edit” undoes up to the last 20 slice-contour edits and two-plane removals. It does not undo seed changes or automatic detection. The contour undo history is not kept after resegmentation or reloading.

Finishing the edit

Return to “Inspect / pan” or close the edit panel, and review the three images and the values. ROI changes are reflected in the measurements, but saving to a file is a separate step. Check the visual-review box again and click “Save results”.

Excluding the internal auditory canal with two-plane contours

What this function removes

An axial contour (left–right × anterior–posterior) and a coronal contour (left–right × superior–inferior) are each extended through the volume, and their intersection is removed. Only the ROI of the selected region is affected. The function does not trace the internal auditory canal automatically.

From drawing contours to removal

1. Open a study that includes MRC, and click “Edit ROIs” → “Remove overlap in two planes”.
2. Choose right vestibule, left vestibule, etc. in “Region to remove”; “Focus selected region” shows the seed area.
3. On the axial MRC, go to a slice where the IAC is easy to see and enclose the area to exclude with at least 3 clicks.
4. On the coronal MRC, also choose a slice and enclose the area, including its superior–inferior extent, with at least 3 points. The last point is joined to the first automatically.
5. Click “Preview removal”. Removal candidates turn yellow, with their voxel count, volume and number of slices.
6. Scroll through both planes and check that the yellow area does not include the vestibule itself; redraw if needed.
7. Click “Remove overlapping ROI” to apply. Measurements are recalculated and the visual-review check is cleared.
8. Click “Close” to return to the three-column view, review, and click “Save results”.

Controls in the editor

Action	Effect
Wheel / slider below each image	Moves the displayed slice in that plane; contours are kept
+ / −, Ctrl + wheel	Zooms both planes at the same mm scale
Tool → Pan image	Drag each image to pan it
Undo last point / Clear contour	Corrects the contour in that plane; preview again
Undo last ROI edit	Also undoes an applied two-plane removal in one step

Avoiding over-removal of the vestibule

A contour is projected at the same position on every slice; a boundary on one slice may not represent the whole internal auditory canal. Start small and, if needed, remove in several steps. With no ROI in the intersection, nothing is removed.

Saving and display orientation

Both planes show physical-scale rulers and voxel dimensions; with large Z spacing, voxels are tall. The ROI after removal and the two contours as applied are recorded in the results ZIP. Unapplied contours, previews and the undo history are not saved. The images are orthogonal planes along the acquisition grid and may remain slightly oblique to true anatomical axial and coronal planes. Resegmentation or automatic detection replaces edited ROIs.

Saving and resuming results

Saving

1. Click “Save results” at the top right and wait until the ZIP has been created.
2. Find the file, such as ELH-Case_XX_….zip, in the browser's download list or Downloads folder.
3. Move it to the folder for that study; do not confuse it with another study with the same serial number or a re-save.

Contents of the results ZIP

File	Contents
ppi.nii / pei.nii / hydrops.nii	Input images and the signed HYDROPS image
mrc.nii / mi2.nii	MRC and HYDROPS-Mi2 (with MRC input)
roi_labels.nii	1 = right cochlea, 2 = right vestibule, 3 = left cochlea, 4 = left vestibule
negative_signal.nii	Mask of negative HYDROPS signal within the ROIs
measurements.csv / lumen.csv	Volumes, ratios, sensitivity analysis and flags by region / LUMEN values (page 21)
lumen_skeleton_bins.csv	Counts per angle for “Compare on a common shape” (1.3.0; page 34)
report.json / README.txt	Method, series information, seed points, measurement details, review status, LUMEN records, etc.
QC_R_…png / QC_L_…png	Three images at the central slice of each side; need seed points and a supported browser
LUMEN_C_… / LUMEN_V_… / LUMEN_NORM_….png	LUMEN_C and LUMEN_V per side (page 21) and, from 1.3.0, LUMEN_NORM_SPIRAL, _STRETCH and _PROFILE per side (page 34). Up to 10 images

Opening saved results

Choose a ZIP saved by this app, as is, with “Open saved results (.zip)” on the left. Images, ROIs, seed points, measurements and the review status (including the display name, notes, cochlear apex and LUMEN steps) are loaded. The zoom, the choice of displayed images and the undo history are not restored.

Not saved automatically

“OPEN STUDIES” lists the studies open in this tab. Before closing or reloading, save each needed study as a ZIP; save again after an ROI change. The distribution ZIP (app) and results ZIP (study data) are separate files.

Loading individual files

Open “Import individual files” and, for each role, choose a single-series DICOM ZIP or a .nii / .nii.gz file. PPI and PEI are required; MRC is optional. After “Create images”, click “Detect & measure” if you need automatic processing. Do not choose a ZIP that mixes several series.

Storing the data

The results ZIP contains images. Even with identifying DICOM tags removed, complete anonymization is not guaranteed. Track the correspondence with the source DICOM within your institution, and store the ZIP like study data.

Troubleshooting and supported scope

Symptoms and what to check

Symptom	Check / action
HTML opens as text / blank page	Extract the ZIP and open it in the latest Google Chrome (page 1). Ask your administrator about browser restrictions in your organization
Series not offered as candidates	Choose the study again, assign series manually by name, TI, etc., and check the format
Coordinate, shape or spacing mismatch	Choose PPI / PEI / MRC from the same study on the same grid. Registration is not implemented and must be done separately
Stops on compression type / number of frames	Re-export the DICOM in a supported format (below)
Automatic detection fails / values look implausible	Check MRC, misregistration, seed points and contamination. Correct seed points or contours and review again (page 14)
LUMEN “Start analysis” cannot be clicked	Check that ① and ② are done for each ear to analyze, or that ears not analyzed are excluded (page 15)
Processing stops / out of memory	Save what you can, close other tabs and reopen. Handle one study at a time
Results cannot be found	Check the browser's download list. Unsaved work in a closed tab cannot be recovered

Supported formats

Single-frame MONOCHROME2 16-bit integer MR DICOM in uncompressed Little / Big Endian or JPEG Lossless (.57 / .70); not JPEG2000, JPEG-LS or Enhanced multi-frame. 3D NIfTI-1 needs an sform and mm units.

Conditions that stop processing, and size limits

Processing stops for different patient IDs, studies or coordinate systems, mixed series, mismatched grids, and unevenly spaced, duplicated or missing slices. Limits: 40 million voxels per volume, 512 MB per file read, and 512 MB and 4,096 entries per ZIP after extraction. No motion correction or automatic resampling between images.

Role and validation of automatic evaluation

A research prototype that matches positions with a reference template averaged from 6 cases of the public dataset VS-SEG and segments by local region growing. It is not a deep-learning model. On other scanners and protocols automatic localization can miss, so check the seed points and place them again manually if they are off (page 14). Independent validation of segmentation accuracy, accuracy on other scanners and protocols, and agreement with manual measurement have not been confirmed.

Scope of distribution

This is the public build for research use. The reference template is an average made from the public dataset VS-SEG (CC BY 4.0) (page 14). Nothing sends images outside. Clicking a link to an original paper opens the external site in the browser.

Sources of the methods

HYDROPS-Mi2 method https://doi.org/10.2463/mrms.2012-0027

Grading by cross-sectional area and morphology https://pubmed.ncbi.nlm.nih.gov/19221900/

Display and unsaved-change settings

Opening the settings

Click “Settings” at the top right. Changes take effect immediately and are stored in the same browser. They may not carry over to another PC or browser, after moving the HTML, or after clearing browser data.

Defaults

Item	Default
ROI transparency	58%
Warn about unsaved changes	ON
Hide potentially identifying information	ON

ROI transparency

0% shows the colors opaque; 100% hides the ROI colors. It applies to the three images (PPI, PEI, HYDROPS, etc.) and the two-plane editor. Yellow removal candidates and contours being drawn stay visible. Image values, ROIs and measurements do not change. The review PNGs in the saved ZIP are written with fixed display settings.

Unsaved-change warning

New image creation, automatic detection, seed and contour edits, two-plane removal, undo and changes to the visual-review status are tracked per study. Changes to the display name and notes, and to the cochlear apex, “Not needed (no contamination)” and exclusion in LUMEN manual analysis, are also tracked. Only opening and viewing saved results does not count as unsaved. The list on the left shows “Unsaved”.

Switching studies and closing

You are asked before switching studies or resegmenting; “Cancel and go back” cancels. Applied results of other studies stay in the page and are checked when closing, as are unapplied contours. Reloading and closing use the browser's standard confirmation, which may not appear in cases such as a forced quit.

Confirming that the ZIP was saved

When ZIP export starts with “Save results”, the applied results at that moment are treated as saved. The browser does not report whether a download finished or was canceled, so check that the ZIP is in the download folder. Unapplied contours are not included in the ZIP.

Hiding identifying information

Free-text fields such as study names, dates, series names and Image Type, and the file names of individually loaded files, are hidden. Patient names and IDs are normally not read or displayed. Turning the setting off restores the original display. It affects the screen only: it does not anonymize saved ZIPs or remove text burned into images.

Copying images and bilateral volume ratios

Copying the displayed image

Click “Copy image” below an image frame. All three frames have it, whichever of PPI, PEI, MRC, HYDROPS or HYDROPS-Mi2 is displayed. The PNG reflects the current slice, zoom, position, contrast, ROI transparency, seed points and any contour being drawn. It includes orientation labels, the image type and the slice number; no heading such as the study name is added. With “Save PNG”, the file name is the image type (e.g., PPI_slice_012.png). Copying the three frames together: page 13.

Pasting an image

1. Display the slice and position you want, and adjust the ROI colors and seed points if needed.
2. Click “Copy image” below the image.
3. In an app that accepts images, such as Word or PowerPoint, press Ctrl+V (⌘V on Mac).

Copying bilateral volume ratios

Click “Copy ratios” next to the heading “Candidate endolymph volume ratios”. The four regions (right cochlea, right vestibule, left cochlea, left vestibule) are copied together. These are volume ratios over the whole 3D candidate region, not the area ratio on the displayed slice.

Copied text

Item	Contents
Volume ratio (%)	One decimal place, as on screen
Negative-signal and ROI volumes (mm³)	Volumes of the numerator and denominator
Not measured	Regions without an ROI or not measured
Visual review	Reviewed / Review required

When copying is not permitted

Browser or PC settings may block direct copying to the clipboard. An image is then shown as a preview: right-click it to copy it, or click “Save PNG”. Text is shown selected in a box, so copy it with Ctrl+C (⌘C on Mac).

Not the same as saving a results ZIP

Copying images or volume ratios does not save your edits. To resume work later, save a ZIP with “Save results”. Copying does not change the measurements or the unsaved state.

Image extent and verification status

Only the image inside the display frame is copied, not parts outside it when zoomed. Text burned into the source images is not removed. Use on an actual Windows PC, direct file:// launch and pasting into specific apps are untested.

Switching between Japanese and English

How to switch

On the welcome screen, choose English at the top right. On the work screen, choose “Settings” at the top right → “Language” → “English”. To return to Japanese, choose “日本語” in the same field.

Switching during work

No reload is needed. The display language changes while the loaded images, current slice and zoom position, ROIs, measurements and unsaved state are kept. The setting is stored in this browser and applies at the next startup. If it cannot be stored, a message says so.

Main buttons

Japanese	English
設定	Settings
DICOMフォルダを選択	Select DICOM folder
自動検出・計測	Detect & measure
抽出範囲を修正	Edit ROIs
2方向で重なりを削除	Remove overlap in two planes
画像をコピー / 体積比をコピー	Copy image / Copy ratios
結果を保存 / 保存結果を開く	Save results / Open saved results
LUMEN の手動解析へ / 解析開始	Go to LUMEN manual analysis / Start analysis

Copied and saved text

“Copy ratios” outputs the four bilateral regions in the selected language. In English the regions read Right/Left cochlea and vestibule, and unmeasured regions read “Not measured”. When you save in English, the region names in the CSV files in the ZIP and the README are also in English. Values, images and ROIs do not change.

Source data and system dialogs

Source data such as DICOM series names are not translated. report.json in the saved ZIP keeps its existing format and the source information. File dialogs, browser permission prompts and reload warnings follow the language of the OS or browser.

The English operation manual

Operation_Manual_EN.pdf (this manual) is the English edition of the Japanese manual, with the same chapter order and page numbers. It is included in the same distribution folder as the Windows BAT file.

Playing the 5-second opening

Visualization and quantification of endolymphatic hydrops on MRI

An image of the inner ear and the ELH Studio title appear for about 5 seconds with a light sweep and fades. When it ends, the welcome screen appears. There is no sound.

Controls

Action	Effect
Start the app	If the setting is ON, plays before the welcome screen
Skip / Esc key	Ends without waiting and goes to the next screen
“Play opening” on the welcome screen	Returns to the welcome screen after playing
“Play opening” in Settings	Returns to the settings screen after playing

Skipping it from next time

Under “Settings” at the top right, turn off “Play the 5-second opening on startup”. The welcome screen can still be shown. The setting is stored in the same browser.

Relation to the welcome screen and OS settings

When “Show welcome screen on startup” is off, the opening does not play automatically either. It is also skipped when the OS is set to reduce motion or animation. Played manually in that state, it shows a still title screen.

Japanese and English

The buttons and the subtitle follow the selected language. In English, the play button reads “Play opening”, the skip button “Skip”, and the startup setting “Play the 5-second opening on startup”.

About the images

The inner-ear image is decorative and is not generated from the loaded study images. Playing or skipping the opening does not change images, ROIs, measurements or the unsaved state. The video material is embedded in the HTML; no external communication or extra video file is needed.

Three-view copy, display name and notes

Copying the three frames together

“Copy all three views” below the image frames (under the operation hints) copies one image with the left, center and right frames side by side as displayed. Each frame has a heading with the image type and slice number, and orientation labels. Quality and extent are as on screen. With “Save PNG”, the file name is HYDROPS_3views_slice_012.png (HYDROPS_3views_slice_012_record.png with record information).

Adding record information

Selecting “Add record information” adds below the image the display name, creation date and app version, the volume ratios of the four bilateral regions as in “Copy ratios”, and the visual-review status. This choice is not saved and is off each time you open the app. An empty display name is written as “No display name”. The study name, study date and notes are not included. The “Hide potentially identifying information” setting does not apply to the display name; the text you entered goes into the image as is. Check it before sharing it outside.

Entering a display name and notes

Open “Name and notes” below the screen heading (study name). “Display name” is one line (up to 80 characters); “Notes” is multiline (up to 2,000 characters). Input is applied when you leave the field, and the study becomes unsaved. Studies with notes show “has notes” in the list on the left.

How the display name is used

State	Shown in the list and heading
Display name empty	“Study NN” as before (the original study name when identifying information is not hidden)
Display name entered	The display name entered; not affected by the “Hide potentially identifying information” setting

Saving and restoring

The ZIP written by “Save results” records them in report.json as label (display name) and memo (notes) and also writes notes.txt. Loading with “Open saved results” restores the display name and notes. ZIPs from 0.7.0 and earlier can also be loaded (with an empty display name and notes).

Caution

- The display name and notes are saved exactly as entered. Do not enter patient names, patient IDs or other identifying information.
- The study name of the source DICOM (title in report.json), images, ROIs and measurements are not changed.
- Input and saving were checked with synthetic cases in Chrome on macOS, and reloading with automated tests. Not checked on an actual Windows PC.

Reference templates and manual seed placement

The reference template is an average of public data

The reference template for automatic localization is one average of 6 cases of the public dataset VS-SEG (vestibular schwannoma, TCIA, CC BY 4.0) and their left-right mirrors, superimposed on the inner-ear center points; it is not the image of any single person. On other scanners and protocols it can place the seed points off target or fail to detect them, so after automatic detection check the four seed points and place them again manually if they are off. Image creation, ROI segmentation and the volume-ratio calculation are the same when seed points are placed manually.

Check the seed points and set them manually if they are off

“Show measurement details and limitations” shows “The reference template is the average of 6 cases of the public dataset VS-SEG. Check the center points after automatic detection. Generalizability to other scanners and protocols is unverified.” If automatic detection was used, turn on “Seed points”, check the positions of the four seed points with “Right inner ear” and “Left inner ear”, and place them again if they are off.

Placing the four seed points manually

1. Click “Edit ROIs”, and under “Tool” choose “Place seed → resegment”. The left image switches to MRC.

2. In “Region to edit”, choose right cochlea, adjust the slice and zoom, and click on the MRC fluid signal of that region.

3. Choose right vestibule, left cochlea and left vestibule and click in the same way. When all four points are placed, the ROIs of the four regions are segmented and measured.

4. Check on every slice of the three images that each ROI stays within the target structure, and correct it with contours if needed (pages 5–6).

Messages about automatic localization

Message on screen	Meaning and action
The inner ear could not be localized with sufficient similarity. Place seed points manually.	Automatic localization failed. Place seed points for all four regions
Automatic localization is inconsistent between sides.	The detections on the two sides disagree. Place seed points for all four regions
Right inner ear: low similarity to the reference. Check the location.	The detected position may be off. Check the seed points and place them again if off (the same for the left inner ear)
Place seed points for all four regions within the image.	Seed points are missing or outside the image

Placing seed points again

Place seed points on the inner-ear fluid signal on MRC (segmentation from seed points requires MRC). Changing a seed point resegments all four regions, so earlier contour edits are replaced. Correct the seed points first, then refine the contours and remove contamination from the internal auditory canal (page 6).

LUMEN manual analysis: steps

What LUMEN is

LUMEN is a research display for viewing the extent of the endolymphatic space as shapes: the cochlea stretched along the centerline of its duct (LUMEN-C) and a map of the rim thickness of the vestibule (LUMEN-V). It requires MRC, the HYDROPS image and the four ROIs. The HYDROPS image, ROIs and volume ratios (measurements.csv) do not change. The values are reference values for research and cannot be used for diagnosis.

Opening it

After checking the three images and ROIs, click “Go to LUMEN manual analysis” below the three-view display. The “LUMEN manual analysis” screen opens and the three views are hidden (“Back to the three views” returns to them). The analysis does not start automatically. On the left are MRC images in two planes (axial and coronal); on the right is the step panel. Switch ears with the “Right” and “Left” tabs; each tab shows the status of ① and ② (Done, Not yet, Not needed, Excluded).

Three steps for each ear

1. ① Remove CSF of the internal auditory canal: choose “Region to remove” and, on both the axial and the coronal MRC, enclose the contaminating part with at least 3 clicks. Check the yellow removal candidates with “Preview removal” and apply with “Remove overlapping ROI” (the same mechanism as page 6). If there is no contamination, check “Not needed (no contamination)”.

2. ② Place the cochlear apex: set “Tool” to “Place cochlear apex” and, on the axial MRC, click the anterolateral tip of the spiral (not the small dot on a slice above). The coordinates are shown; “Clear” removes the apex.

3. ③ Start analysis: “Start analysis” becomes available once ① and ② are done for every ear to analyze (every ear not excluded). The step panel shows the status of ① and ② for each ear.

About the cochlear apex

The cochlear apex sets the up/down direction of LUMEN-C for that ear (top = scala vestibuli side) and the initial axis of the spiral fit. It is shown as a blue-green circle on the axial and coronal images. If the apex is more than 12 mm from the cochlear seed point, or less than 1.0 mm from the fluid centroid, LUMEN-C of that ear ends in an error. Check the apex against the resulting stretched view and the centerline display; the “Deviation between apex and axis” value cannot reveal a misplaced apex. The warnings on the winding direction (from 1.2.2) and on up/down against the skeleton (from 1.2.4) also help (pages 29–31).

LUMEN manual analysis: exclusion, progress and redoing

Ears not analyzed (exclusion)

For an ear that cannot be evaluated, for example because of a cochlear implant or because it is outside the imaging range, check “Do not analyse this ear” and choose a reason. The result frame of an excluded ear shows “Not analysed” and the reason, and saving writes an exclusion row instead of results to lumen.csv. With one ear excluded, “Start analysis” works with the other ear only. Clearing the exclusion restores that ear's results (if the ROI is the same).

Exclusion reasons

Option on screen	excluded_reason in lumen.csv
Cochlear implant / metal artefact	implant
IAC contamination cannot be removed	csf_not_removable
Outside the imaging range	out_of_fov
Other (enter a reason of up to 80 characters)	other

Progress display

Clicking “Start analysis” shows a spinning mark and “Analysing the right ear (1/2)…” to the right of the button, and the button reads “Analysing…”. Each ear takes a few seconds. When both ears are done, the display disappears and the results appear below, right then left.

The screen after the results

When results appear, the two-plane MRC and the step panel are collapsed, and a bar “The two-plane MRC and the step panel are collapsed” appears above. A study with results also opens collapsed.

Redoing

To correct contours or the cochlear apex, click “Fix contours / apex and analyse again”; the editing area and the step panel return. Results after an ROI change show “Outdated result (analyse again)”, so click “Start analysis” again. “Undo last ROI edit” in the step panel reverts the last ROI edit whichever ear is selected (it also works on the tab of an excluded ear).

Saving and resuming

The results go into the ZIP with “Save results” (page 21). When saved results are opened, ears with ① and ② done show “The saved results include LUMEN results. Press “Start analysis” to show them (the numbers will be the same).” If there is an unapplied contour when you click “Back to the three views”, you are asked to confirm.

Reading LUMEN results: summary section

Layout of the results screen

A summary section is at the top of the results screen, followed by the right and left result frames. Each frame shows the HYDROPS axial viewer (page 19), LUMEN-C and LUMEN-V (page 18), the section “Compare on a common shape” (1.3.0; page 32), and the table of values and warnings (page 20). The values are reference values for research; no thresholds or normal ranges have been established.

Two values per ear

Value	Meaning
Basal-turn endolymph fraction	Proportion (%) of endolymph (negative on HYDROPS) in the fluid inside the duct within 1.6 mm of the centerline over the first cochlear turn (from the point where the fluid starts to 360°). It combines the scala vestibuli and scala tympani sides and does not depend on the up/down judgment
Wall-contact fraction	Proportion (%) of the vestibular wall area touched by endolymph: the share of directions, excluding openings, where the perilymph rim is thinner than 0.3 mm, weighted by solid angle (formerly “Rim-loss area”)

Bars, vertical lines and side difference

Each value is a 0–100% bar, with the other ear's value as a vertical line (legend “Vertical line: the other ear's value”). The right column shows the side difference, such as “Right − left −45.4 points” (positive if the right is larger; the number is an example). The small line under each name gives the counted range: “Cochlea: within 1.6 mm of the centerline, 0–360°” and “Vestibule: openings excluded, rim thinner than 0.3 mm”. Excluded ears show “Excluded (reason)”.

Scala vestibuli and scala tympani side values (under verification)

For ears with a cochlear apex, E_SV / E_ST of the basal turn (endolymph occupancy on the scala vestibuli side and on the scala tympani side) are also shown; manual analysis requires the apex, so they are always shown. Which side is the scala vestibuli side agreed with the ground truth in the public IE-Map data, but real images have no ground truth and it has not been confirmed, so they carry the mark “(under verification)”.

Warnings and outdated results

If an ear has warnings, “Warnings present (see the panel below)” appears; check the warnings in the frame below before reading the values. If the result of either ear is “Outdated result (analyse again)”, the vertical lines and the right − left difference are not shown.

Difference from the volume ratios on the three-image screen

“Volume ratios from the three-image screen (HYDROPS)” in the same section lists “Cochlea %EH (whole ROI)”, “Vestibule %EH (whole ROI)”, the negative-signal / total volume (mm³) and “Mi2 …% (reference)” (always the current values). %EH is the volume ratio over the whole ROI and counts a different range from the basal-turn endolymph fraction, so the two are not the same value.

Reading LUMEN results: stretched view and map

LUMEN-C: stretched cochlea view

The spiral cochlea is laid out as a single band along the centerline of the cochlear duct obtained from MRC. The three rows are, from the top, MRC, HYDROPS and the overlay. Horizontally is the position along the centerline, with dotted ticks every 90°. Vertically is the direction across the duct: the top is the scala vestibuli side and the bottom the scala tympani side. Both axes have the same scale, with 5 mm and 1 mm scale bars below.

The overlay row and the range to read

In the overlay row, red is the endolymph side, cyan the perilymph side and black outside the fluid. In hydrops the cochlear duct bulges toward the scala vestibuli side, so endolymph (red) is expected to spread upward. If endolymph is abundant both above and below, suspect insufficient enhancement or CSF contamination rather than hydrops. Read mainly the basal turn (0–360°); beyond the middle turn the centerline is unstable, so it is for reference only.

LUMEN-V: map of the vestibular rim thickness

Imagine standing at the center of the vestibule and looking around. Lines are cast from the center in all directions, and the thickness of the perilymph layer from the outer edge of the endolymph-filled sacs (utricle and saccule) to the bony wall is color-coded per direction on one map. The labels below the map (Posterior, Medial, Anterior, Lateral, Posterior) give the horizontal direction, and those on the left (Superior, Horizontal, Inferior) the vertical. The left and right edges are both posterior and connect.

Map colors

Color	Meaning
Bright (yellow to white)	Thick perilymph layer; the sac is away from the wall
Dark (purple)	Thin perilymph layer
Black	No perilymph layer; the sac touches the wall
Gray (opening)	Openings to the semicircular canals and the cochlea; no wall, so not measurable

Origin of the angle

Angle 0° is not the basal end of the centerline but the point where the fluid starts (the first point assigned fluid within 1.6 mm of the centerline). The centerline fit starts from the vestibular side, so in some ears its beginning lies outside the cochlear fluid. The angle θ0 from the end to that point is kept as theta0_deg in lumen.csv.

Cautions when reading

Even in normal ears, the areas where the utricle and saccule attach to the wall are black. A larger black area is read as the sacs approaching the wall, but black alone does not mean an abnormality. Near the top and bottom edges the map is stretched horizontally and looks larger than it is. The map orientation is approximate, based on the image axes. Do not judge the up/down direction from the HYDROPS or overlay rows of the stretched view (that would treat the side with endolymph as the scala vestibuli side and bias E_SV upward).

Reading LUMEN results: comparison with the HYDROPS axial image

HYDROPS axial viewer

For each ear with results, a HYDROPS axial viewer appears above the stretched view, with the cochlear centerline (white; brighter near the displayed slice), ticks every 90° (yellow) and the cochlear apex mark (blue-green) overlaid. “Image” selects HYDROPS, HYDROPS-Mi2 or MRC. The right and left viewers move independently.

Viewer controls

Action	Effect
To cochlea / Fit view	Back to the initial zoom position / full view
− / +, Ctrl + wheel	Zoom out / in (also ⌘ + wheel on Mac)
Slider / wheel / drag	Change the slice (on the thin plane, the depth) / pan
Clear the mark / Esc key	Clears the pinned mark

Linking with the stretched view

Hovering over the stretched view marks that point on the axial image in yellow, and a yellow line shows the angle, the distance along the centerline, the distance toward the scala vestibuli or scala tympani side, and the slice (the slice does not move). Clicking pins the point and moves the slice to it. Clicking near the centerline (within 1.6 mm) on the axial image also pins the nearest point. Before the point where the fluid starts, the angle is negative (e.g., -20°).

Switching the “Plane”

Plane	Appearance
Axial	The acquired axial slices (as before)
Cochlear-axis view, thin plane	Viewed from the cochlear apex side of the skeleton (0° to the right; ears where the fit failed are viewed along the spiral axis from the apical side; page 29). The slider or wheel changes the depth (“Depth … mm from the apical end”, in 0.2 mm steps)
Cochlear-axis view, projection	The whole cochlear height in one image; no depth control. Red marks pixels with negative signal in the ROI somewhere along the depth, not an amount

Show unexplained fluid (magenta)

When checked, the voxels counted by the “Unexplained fluid” warning (fluid in the cochlear ROI more than 1.6 mm from the centerline) are overlaid as magenta squares: on the axial view those on the displayed slice, on the thin plane those near the depth, and on the projection all of them.

About the planes

The labels at the edges of a plane (Lateral, Medial, Anterior, Posterior, Superior, Inferior) are approximate directions. “Depth 0 mm from the apical end” is the upper end of the plane range, not the position of the apex mark itself. The plane and image choices are not saved.

Reading LUMEN results: table of values and warnings

Cochlea (LUMEN-C) rows

Row name	Meaning
Basal-turn endolymph fraction	The same value as in the summary section (%)
Upper-turn endolymph fraction (reference)	The same value beyond the basal turn (angle above 360°); for reference, as the centerline is unstable beyond the middle turn
Basal minus upper (reference)	Basal-turn endolymph fraction − upper-turn endolymph fraction (points)
Basal turn (0-360 deg) E_SV, E_ST	Endolymph occupancy of the basal turn on the scala vestibuli and scala tympani sides (0–1; under verification)
Perilymph ratio SV/ST (basal turn, reference, under verification)	Perilymph pixels on the scala vestibuli side ÷ those on the scala tympani side in the basal turn. Not in the summary
Centerline length, Turn angle	Centerline length (mm) and turn angle from the point where the fluid starts (°)
Endolymph occupancy, scala vestibuli / scala tympani side (E_SV, E_ST) (reference)	Whole-cochlea values (under verification). Values per 90° section (reference) follow at the end of the table
Unexplained fluid	Proportion (%) of fluid in the cochlear ROI more than 1.6 mm from the centerline
Orientation conflict, Up/down swap	In manual analysis the apex decides up/down, so “—” and “No”
Cochlear apex specified, Deviation between apex and axis	Whether an apex is set, and the angle (°) between the fitted axis and the apex direction; about 15–21° even when placed correctly
CSF removal	Status of ① (Done, Not needed, Not yet)

Vestibule (LUMEN-V) rows

Row name	Meaning
Median rim thickness	Median rim thickness (mm; weighted by solid angle)
Wall-contact fraction	The same value as in the summary section (%)
Wall contact, upper half / lower half (reference)	Calculated only over the half above / below latitude 0
Number of openings, Openings reached by endolymph	The number of openings and how many of them the endolymph reaches

Warnings

Warnings appear below the result frame: much fluid far from the centerline (unexplained fluid above 10%; possible IAC/CSF contamination or a poor centerline fit), an axis 30° or more off the apex direction, many vestibular openings, a centerline longer or shorter than usual, a large residual of the spiral fit, and others. For an ear with warnings, check the mask, stretched view and centerline; remove any contamination in ① and analyze again. Orientation and skeleton warnings and rows: pages 29–31; the rows of “Compare on a common shape” and the reliability warning (1.3.0): page 33.

Saved LUMEN records

File name

The name of the “Save results” ZIP includes the save time (e.g., ELH-Case_01_20261020-0905.zip). When “Hide potentially identifying information” is off, the study title replaces the serial number. ZIPs saved by earlier versions (HYDROPS-Case_… etc.) can also be loaded as is with “Open saved results”.

lumen.csv columns (2 rows per ear, cochlea and vestibule; 29 of 45; rest: pages 28, 31, 30, 34)

Column	Contents
side, structure	Ear (R / L) and structure (cochlea / vestibule)
length_mm, turns_deg	Centerline length (mm) and turn angle (degrees). Cochlea row
E_SV, E_ST, basal_E_SV, basal_E_ST, basal_P_SV_over_P_ST	Endolymph occupancy on the scala vestibuli / scala tympani sides (0–1; whole and basal turn) and the perilymph ratio SV/ST. Under verification
unexplained_fluid_pct	Proportion of unexplained fluid (%)
rim_median_mm, rim_loss_area_pct, n_openings, openings_with_endolymph	Median rim thickness, wall-contact fraction (%), number of openings, openings reached by endolymph. Vestibule row
orientation_conflict, flipped, apex_set, warnings, studio_version	Orientation conflict, up/down swap, apex set, warning names (unexplained_fluid, etc.), app version used
csf_removal, excluded_reason	Status of ① (done / not_needed / empty) and the exclusion reason
basal_EL_pct, upper_EL_pct, basal_minus_upper_pct, basal_n_voxels, upper_n_voxels	Basal and upper-turn endolymph fractions (%), their difference (points) and fluid pixel counts. Cochlea row
rim_loss_upper_pct, rim_loss_lower_pct	Wall-contact fractions of the upper and lower halves (%). Vestibule row
theta0_deg	Origin of the angle θ0 (degrees). Cochlea row

What goes into report.json

lumen holds each ear's record (indices, warnings, computing version, etc.), lumen_manual each ear's “Not needed” marks and exclusion, and lumen_saved_sides the ears whose saved LUMEN records match the current ROIs. The cochlear apex (apex) is also saved. measurements.csv (volume ratios) is the same with or without LUMEN.

LUMEN images (1.2.5; “Compare on a common shape” images: page 34)

“Save results” adds LUMEN-C (MRC, HYDROPS, overlay) and LUMEN-V images of analyzed ears, except excluded ears, failed structures and ears without records. As on screen, LUMEN-V colors are fixed at 0–1.6 mm, and LUMEN-C MRC and HYDROPS rows are auto-windowed per ear (image), so do not compare them across cases. In the overlay, red/cyan (sign) and distribution are comparable; intensity is auto-adjusted too. Header: display name, ear and type, summary values, warnings, input type and enhancement QC, creation date and version (no study date, patient ID, title or notes). Re-saving without reanalysis keeps them (a new display name redraws only the header); after a language change or for ZIPs up to 1.2.4, save after “Start analysis”. Record: report.json lumen_images.

Empty columns and which ears are written

If a ZIP saved by an earlier version is saved again without reanalysis, the 9 columns from basal_EL_pct to theta0_deg are empty (“Start analysis” fills them). After an ROI edit, only reanalyzed and excluded ears are written.

Research use and limitations

Not for diagnosis

ELH Studio is a research prototype, not for diagnosis. All values, including volume ratios and LUMEN summary indices, are reference values for research. No thresholds or normal ranges have been established.

Current limitations

Item	Status
Thresholds and normal ranges	Not established. The distribution of values in real cases has not been evaluated
Scala vestibuli / scala tympani side judgment	“under verification”. Agreed with the ground truth in the public IE-Map data (one ear and its mirror image, 9 conditions), but real images have no ground truth and it has not been confirmed
Beyond the middle turn	Reference values, as the centerline is unstable
Studies with insufficient enhancement	Where the contrast agent does not reach the perilymph sufficiently (PPI darker than PEI), %EH and LUMEN values are overestimated. Automatic judgment of enhancement sufficiency was planned for the next version (added in 1.1.0 as Enhancement QC; pages 23–25)
Automatic localization	Reference template: an average of 6 cases of the public dataset VS-SEG. It can miss on other scanners and protocols; check the seed points and ROIs on the images (page 14)
Input types B and C (1.2.0)	Values from PPI and MRC (HYDROPS2) and from a single HYDROPS-type image are separate measurements from PPI, PEI and MRC; do not compare values or thresholds. The single HYDROPS-type image is unverified with actual 3D-real IR data (pages 26–28)
Cochlear orientation and skeleton (1.2.2, 1.2.4)	Warning thresholds (axis direction 45°; distance from the skeleton 2.2 mm, total 90°) are provisional. How well the skeleton fits on other scanners and protocols is unconfirmed (pages 29–31)
Compare on a common shape (1.3.0)	The tube thickness of 1.5 mm and the criteria of normalization reliability are provisional. The relation of the endolymph fraction in the tube to the degree of hydrops, normal ranges, and normalization on other scanners and protocols are unconfirmed (pages 32–34)
Scope of verification	LUMEN calculations were checked with analytical models and the public IE-Map data (simulated HYDROPS), and screen behavior with synthetic cases only (the public IE-Map and VS-SEG data were also used for the orientation and skeleton of 1.2.2 / 1.2.4 and the normalization of 1.3.0). Agreement between the JavaScript and Python reference implementations shows that the two agree, not that they are correct. Accuracy in real cases and on other scanners and protocols, and agreement with manual measurement, are unconfirmed
Operating environment	Google Chrome only (other browsers are not supported; page 1). Screens were checked in Chrome on macOS (headless). Launching on an actual Windows PC (BAT, direct file:// launch) is unverified

Before reading the values

Check the warnings, the stretched-view overlay, and the up/down direction and contamination on the axial and cochlear-axis planes. As with volume ratios, visually review the ROIs and IAC contamination before reading the values.

Enhancement QC (sufficiency of perilymph enhancement)

What it judges

The HYDROPS method assumes that enough contrast agent reaches the perilymph and that the perilymph is bright on PPI. If not, PPI becomes darker than PEI, and %EH and LUMEN values are overestimated. Enhancement QC judges the sufficiency of enhancement before measurement and stops measurement if it may be insufficient. It does not change ROIs or %EH.

Rules (two stages)

Item	Contents
Stage 1: anterior chamber	Finds the globes on MRC. Insufficient if the “Anterior chamber PPI/PEI” (mean PPI ÷ mean PEI) within 4 mm of the front surface is below “Minimum PPI/PEI” (default 1.0)
Stage 2: four inner-ear regions	For each ROI (right and left cochlea and vestibule), “below threshold” if PPI/PEI is below the minimum (default 1.0) or “PPI/reference” is below “Minimum PPI/reference” (default 1.5)
“Reference (pons PPI signal)”	Median PPI of non-fluid pixels in a sphere of radius 15 mm, 10 mm posterior to the centroid of the seed points. With fewer than 500 non-fluid pixels, “No reference” is shown and the judgment is made without PPI/reference
Side rule	A side is insufficient only when both its cochlea and vestibule are below threshold (in hydrops, a single region can be low)
Overall verdict	“Possibly insufficient enhancement” if the anterior chamber or both sides are insufficient, or one side is and the other cannot be judged. One side only: “Possibly insufficient enhancement (one side only; measurements continue; severe hydrops or ROI contamination can also cause this)”. No MRC or PEI, or neither globes nor ROIs: “Enhancement QC: undetermined”. Otherwise “Enhancement QC: sufficient” (with the 1.2.0 input type “PPI and MRC (HYDROPS2)”, judged without PEI by PPI/reference only; page 27)

When it judges and what it shows

It re-judges at each “Detect & measure”, resegmentation, ROI edit, apex placement, opening of saved results and threshold change. When sufficient, a line such as “Enhancement QC: sufficient (anterior chamber PPI/PEI 2.15, pons reference 12.0)” appears below the measurement cards (example numbers). With four seed points but no ROI pixels, only the anterior chamber is judged, and the line ends with “(inner-ear values after ROI extraction)”.

When it cannot judge, or the globes cannot be evaluated

Without MRC or seed points, it does not stop and gives the reason, such as “Enhancement QC: undetermined (MRC is missing)”. If the globes are not found, “The globes could not be evaluated (outside the field of view, artefact or signal loss). The verdict uses the inner-ear values only.” appears, and only stage 2 is used.

Threshold settings

Under “Enhancement QC thresholds” in “Settings” at the top right, you can change “Minimum PPI/PEI” (0.5–2.0, default 1.0) and “Minimum PPI/reference” (1.0–3.0, default 1.5) in steps of 0.05. A change redoes the judgment of the open studies, and the thresholds used are recorded in report.json. QC cannot be disabled.

Enhancement QC: when insufficient, and records

When “Possibly insufficient enhancement” appears

1. The “Enhancement QC” panel appears instead of the measurement cards. Until you continue, “Save results”, “Go to LUMEN manual analysis”, “Copy ratios”, “Copy image” and “Copy all three views” are disabled.

2. Check “Anterior chamber PPI/PEI” and “Reference (pons PPI signal)” in the panel, and “PPI/PEI” and “PPI/reference” for each region. Values below a threshold are marked “below threshold”. “Vitreous PPI/PEI (reference)” is the value inside the globe, shown for reference only and not used in the judgment.

3. “Go to the globe slice” (only when the globes were found) and “Go to the vestibule slice” move the three images there; compare the brightness of the front of the globe and the vestibule on PPI and PEI. If PPI is darker than PEI in the anterior chamber, enhancement is insufficient. When only the inner ear is dark, severe hydrops can also cause it.

4. If contamination by the internal auditory canal or CSF seems to be the cause, edit the ROIs (pages 5–6; editing remains available while stopped). The judgment is redone after each edit, and the measurement cards appear once it becomes sufficient.

5. To continue after reviewing the images, enter a reason in “Note (up to 80 characters, optional)” if needed, and click “I have reviewed the images; continue and record this”.

After continuing

A bar “Continued despite the insufficient-enhancement verdict (recorded)” appears above the measurement cards. It can be closed with ×, and does not reappear after the study is closed or the saved results are reopened. “Enhancement QC: insufficient (continued, recorded)” below the cards remains. The continue record is not removed by ROI edits or by the judgment changing to sufficient. Continuing makes the study “Unsaved”.

What is recorded

File	Contents
report.json	enhancement_qc: verdict, reason codes (EYE_LOW, SIDE_LOW_R / L, EYE_NOT_FOUND, NO_REF, NO_ROI, NO_SEEDS, NO_MRC, etc.), thresholds, reference value, globes, values per region, verdict per side, the continue record (time, ROI revision, verdict, note) and the judgment history (latest 20)
measurements.csv	In the flags column: ENH_LOW_SIDE on the rows of an insufficient side, ENH_LOW_EYE on all rows when the anterior chamber is insufficient, and ENH_QC_OVERRIDE on all rows when continued. Sufficient or undetermined without continuing: unchanged
README.txt	One line at the end: “Enhancement QC: sufficient”, “Enhancement QC: insufficient (continued, recorded)”, “Enhancement QC: insufficient (one side only)”, “Enhancement QC: undetermined”, etc.
Copied records	In the text of “Copy ratios” and in the record of “Add record information” of “Copy all three views”, the same line as README.txt, after the visual-review line

When saved results are opened

If there is a record, it is restored with the continue record, and the bar does not appear. ZIPs saved by earlier versions are judged when opened and stop at that point if insufficient. Batch processing (tools/studio_batch.cjs) does not stop.

Enhancement QC: one side only, LUMEN screen, limitations

When only one side is insufficient

Measurement does not stop. Below the measurement cards appear “Possibly insufficient enhancement (one side only; measurements continue; severe hydrops or ROI contamination can also cause this)” and “Low side: Right” (or Left), and the same text heads the list under “Show measurement details and limitations”. Insufficiency on one side only rarely comes from the acquisition and may come from ROI contamination or pathology, so check the images and ROIs of that side. If one side is insufficient and the other cannot be judged (no ROI, etc.), it is not treated as one side only: measurement stops with “Possibly insufficient enhancement”, and the panel shows “The other side could not be judged (e.g. no ROI)”.

On the LUMEN screen

If the verdict changes to “Possibly insufficient enhancement” while the LUMEN manual analysis screen is open (after an ROI edit or a threshold change), “Start analysis” and “Save results” are disabled and “Stopped by the enhancement QC. Go back to the three-view screen to review the verdict” appears. For continued studies and studies with one side insufficient, the LUMEN summary section begins with the same line as README.txt (“Enhancement QC: insufficient (continued, recorded)”, “Enhancement QC: insufficient (one side only)”).

Limitations

The thresholds are provisional and unverified. Stage 1 cannot be evaluated in studies where the globes are outside the field of view, etc. Inputs without PEI were not supported (up to 1.1.0) (with the 1.2.0 input type “PPI and MRC (HYDROPS2)”, the judgment uses PPI/reference only, and with “One HYDROPS-type image and MRC (scanner subtraction, real IR)” it cannot judge; page 27). The judgment does not correct insufficient enhancement and is not a diagnostic criterion.

Distinguishing hydrops from insufficient enhancement

In severe hydrops the endolymph expands, so PPI can be darker than PEI in the inner ear. Judge together with the anterior chamber PPI/PEI. Flags such as ENH_LOW_SIDE also appear with severe hydrops on one side, so do not use them as criteria for excluding cases.

Input types: PPI and MRC (HYDROPS2), and one HYDROPS-type image

What to choose

For studies without PEI, or with only one signed image made by the scanner, you can choose the “Input type” when creating images (1.2.0). The default is the existing “PPI, PEI and MRC (HYDROPS)”; with the default, the results are the same as in 1.1.0. For every type, ROI segmentation uses MRC only, and the volume ratio counts the negative pixels within the ROI on the HYDROPS image.

Three input types

Input type	Images used, calculation and when to use
“PPI, PEI and MRC (HYDROPS)” (A below; default)	HYDROPS = PPI − PEI, as before
“PPI and MRC (HYDROPS2)” (B below)	HYDROPS2 = PPI − k × MRC (k default 0.04) is used as the HYDROPS image; the Mi2 counterpart is HYDROPS2 × MRC. Requires PPI and MRC; PEI is not used. For studies without PEI
“One HYDROPS-type image and MRC (scanner subtraction, real IR)” (C below)	A signed image made by the scanner (negative = endolymph) is used as the HYDROPS image as is; the Mi2 counterpart is input image × MRC. Requires this image (“HYDROPS-type image”) and MRC. For measuring on the scanner's subtraction image

How to choose

1. Choose it in “Input type” at the top of the series selection screen opened by “Select DICOM folder”. “Import individual files” has the same choice.

2. Choosing B hides the PEI row and shows the “Coefficient k” field. Choosing C hides the PPI and PEI rows and shows the “HYDROPS-type image” row and “Flip the sign (so that negative = endolymph)”. MRC is required for B and C.

3. If there is exactly one candidate for C (a series whose name contains SUB, HYDROPS, REAL IR or PSIR, or whose Rescale Intercept is negative), it is entered in “HYDROPS-type image” automatically. Check it in the header list.

4. Click “Create images from selected series” (“Create images” for individual files). When the page opens, A is always selected; it never switches automatically from series names, etc.

Coefficient k and its effect (B)

The default 0.04 is the value for making HYDROPS2-type subtraction images on the one scanner used in development and has not been verified on other scanners or imaging conditions. The range is 0.010–0.100 (field step 0.005); a blank or out-of-range value stops with “k must be between 0.010 and 0.100.” %EH is sensitive to k. No single k reproduces classic HYDROPS, so do not tune k to match appearances. The 99th percentile of MRC (k_context.mrc_p99 in report.json) is not a value for converting k but a record for review.

Input types: checking the sign, the screen, enhancement QC

Flipping and checking the sign (C)

With C, negative pixels of the input image are counted as endolymph. For an image in which endolymph is positive (bright), load it with “Flip the sign (so that negative = endolymph)” checked. After loading, check that the red overlay (“Negative signal in ROI (candidate endolymph)”) matches the endolymphatic space and that perilymph and bone are not red. If it is reversed, change the sign flip and load again. An image with no negative pixels stops with “There are no negative voxels. Choose a signed image (negative = endolymph) or flip the sign.”

Screen display

The default three frames are PPI / MRC / HYDROPS for B (the HYDROPS frame description includes k, as in “HYDROPS2 = PPI − 0.040 × MRC”) and MRC / HYDROPS-Mi2 / HYDROPS for C (description “Input image (negative = endolymph)”). The formula at the lower left becomes “PPI − 0.040 × MRC = HYDROPS2” (B) or “Input image = HYDROPS” (C), and the cautions for B and C head the list under “Show measurement details and limitations”. “PPI − PEI = HYDROPS” on the welcome and opening screens does not change.

Scale of LUMEN-V

LUMEN-V (vestibular rim thickness, wall-contact fraction) sets its threshold from the magnitude of the positive pixels of the HYDROPS image, so with B and C its values are not on the same scale as with A. The LUMEN summary section begins with “LUMEN-V rim thickness depends on the scale of the input image. With inputs other than PPI − PEI, the values are not on the same scale as the PPI, PEI and MRC (HYDROPS) input.” LUMEN-C (cochlea) uses only the sign and is not affected.

Enhancement QC

With B there is no PEI, so the anterior chamber is not evaluated, and the judgment uses PPI/reference of the four inner-ear regions and the side rule. When sufficient, it reads like “Enhancement QC: sufficient (anterior chamber not evaluable, pons reference 154.1)” (an example from a synthetic case). In the panel, the “Anterior chamber PPI/PEI” row shows “Not evaluated (no PEI)”, and the guidance reads “Without PEI, the anterior chamber and PPI/PEI are not evaluated. Check on PPI that the perilymph of the cochlea and vestibule is brighter than the pons. When only the inner ear is dark, severe hydrops can also cause it.” With C there is no PPI, so “Enhancement QC: undetermined (PPI is missing)” appears and measurement does not stop.

What was checked, and grid mismatches

The behavior of C was checked with synthetic cases; it has not been verified with actual 3D-real IR data (page 28). An image on a different grid from MRC stops with “Image dimensions, position, orientation or spacing do not match. Register the images before loading.”

Input types: records, differences from A, limitations

What is recorded

File	Contents
report.json	input_mode (A ppi_pei, B ppi_mrc, C given), k and k_context.mrc_p99 (B; the 99th percentile of MRC at loading, not a value for converting k but a record for review), sign_flipped (C), formulas (the formulas of each type)
Images in the saved ZIP	B: ppi.nii, mrc.nii, hydrops.nii, mi2.nii; C: mrc.nii, hydrops.nii, mi2.nii (no pei.nii in either; for C, the image after flipping). QC images: QC_R_PPI_MRC_HYDROPS.png for B and QC_R_MRC_MI2_HYDROPS.png for C (QC_L_… for the left)
README.txt	One line after the enhancement QC line. B: “Input type: PPI − k × MRC (HYDROPS2, k = 0.040). Open with ELH Studio 1.2 or later.”; C: whether the sign was flipped, and that real IR is unverified
Copied records	In the text of “Copy ratios” and the record of “Copy all three views”, one line after the formula line, such as “Input type: PPI − k × MRC (HYDROPS2, k = 0.040)” (B)
lumen.csv	input_mode in column 30 (ppi_pei for A studies)

Differences from A

Values from B and C are separate measurements from A; do not compare them with A values or thresholds. The columns of measurements.csv are the same for every type, with no input-type column; check the input type in report.json and README.txt. When A is chosen, measurements.csv, README.txt, the QC image names and the existing items of report.json are the same as in 1.1.0.

Opening saved results

A ZIP of B recalculates HYDROPS2 from ppi.nii, mrc.nii and k in report.json; a ZIP of C uses hydrops.nii as is (it is not flipped twice). Earlier ZIPs without input_mode open as A. ZIPs of B and C cannot be opened in 1.1.0 or earlier versions.

Limitations

C has only been checked for behavior with synthetic cases and has not been verified with actual 3D-real IR data (sign, zero point and bone values unconfirmed). No resampling or registration is done, so prepare the input images on the same grid. In studies with weak enhancement, %EH is overestimated with B and C as with A.

LUMEN: cochlear skeleton and “Cochlear-axis view” images

What the skeleton is (1.2.4)

The skeleton is a predefined cochlear shape: the centerline of the cochlear duct of the right ear in the public IE-Map data, and its mirror image for left ears. During LUMEN analysis it is fitted to the fluid of each ear by rotation, translation and scaling (0.8–1.2×); it is not fitted to the traced centerline. The winding direction is fixed by the skeleton (counterclockwise for right ears, clockwise for left ears). Computed values (endolymph fractions, θ, the stretched view, measurements.csv) are the same as in 1.2.3.

Image orientation

For ears where the skeleton was fitted, the “Cochlear-axis view, thin plane” and “Cochlear-axis view, projection” images of “Plane” (page 19) are always viewed from the cochlear apex side of the skeleton (0° to the right). Right ears always appear counterclockwise and left ears clockwise. Ears where the fit failed keep the earlier orientation, with “The skeleton could not be fitted” at the upper left of the image.

What the image shows

Display	Meaning
Green dashed line	The skeleton (legend “Skeleton (predefined shape)”)
Centerline colors	The traced centerline, colored by turn: cyan (basal turn 0–360°), yellow (middle turn 360–720°), orange (apical turn from 720°). White before the point where the fluid starts
Red sections	Sections off the skeleton (legend “Section off the skeleton”). Only with the departure warning
Angle labels	The labels of the 90° ticks, grouped in one place per direction outside the spiral (e.g., 0° / 360° / 720°) and joined to the ticks by dotted lines. If there is no room outside the spiral, they are placed on top of the image
Upper left / lower left	Upper left: “Right ear: counterclockwise” (“Left ear: clockwise” for left ears) and “(fixed by the skeleton)”. Lower left: the legend (moved up slightly if it overlaps the direction label on the bottom edge)
Red frame at the top of the image	A skeleton warning (page 30)

Criteria and source

The criteria (2.2 mm, total 90°) are provisional, and their validity has not been verified. How well the skeleton fits on other scanners and protocols is unconfirmed. The skeleton comes from IE-Map (CC BY 4.0; Ahmadi SA et al. Sci Rep 11, 3293 (2021)), listed in THIRD_PARTY_NOTICES.txt.

LUMEN: skeleton warnings, table rows and records

Skeleton warnings (1.2.4; below the result frame and at the top of the image)

Start of the warning	Condition, and display at the top of the image
“The traced cochlear centerline departs from the predefined cochlear shape (the skeleton)”	Sections more than 2.2 mm from the same turn of the skeleton total 90° or more. Those sections are drawn in red, with “Warning: from about …° on, the tracking departs from the skeleton”
“The computed up/down is opposite to the apex of the predefined cochlear shape (the skeleton)”	The computed up/down and the skeleton axis are more than 90° apart. “Up/down reversed: the side opposite the skeleton's apex is computed as up (scala vestibuli side)” is shown

Rows added to the results table (cochlea)

Row name	Display
Departure from the skeleton	No / from about …° / Fit failed. “—” for records saved by versions before the skeleton
Largest distance from the skeleton	The largest distance (mm) between the centerline and the same turn of the skeleton
Up/down (against the skeleton)	Correct / Reversed (check) / Not judged

Columns added at the end of lumen.csv (columns 35–38, cochlea rows only)

Column	Contents
skeleton_fit	Proportion of skeleton points inside the fluid (below 0.6: fit failed)
skeleton_dev_max_mm, skeleton_out_from_deg	Largest distance (mm) and the angle where the departure starts (degrees)
skeleton_axis_angle_deg	Angle between the computed axis and the skeleton axis (degrees)

When a warning appears

For an ear with the up/down reversed warning, do not use the scala vestibuli / scala tympani side values until you correct the apex position and analyze again. With the departure warning, do not use the scala vestibuli / scala tympani side values of the departing section. Check the seed and apex positions and contamination; correct contamination with “Remove overlap in two planes” and analyze again. If you open a ZIP saved by an earlier version and save it without reanalysis, these columns are empty.

LUMEN: cochlear orientation warnings (1.2.2)

Checking the cochlear orientation

After the cochlear analysis of LUMEN, the orientation of the fitted cochlea is checked in two ways, and a warning is given if it is off. These are warnings only; the values and the way up/down is decided do not change.

Orientation warnings (below the result frame)

Start of the warning	Condition
“The winding of the cochlea is the reverse of this ear's anatomy”	Followed from base to apex, the centerline winding seen from the computed apex side is not counterclockwise for a right ear or clockwise for a left ear. Up/down may be reversed (this also occurs when the apex position is off). Not judged when the centerline turns less than half a turn (180°) or is almost straight
“The fitted cochlear axis deviates by more than 45°”	Relative to the head orientation built from the four seed points, the fitted axis deviates more than 45° from the typical direction (about 45° lateral-anterior) (45° is provisional). Not judged when the head orientation cannot be built, such as when the two cochleae are closer than 40 mm

Rows added to the results table (cochlea)

Row name	Display
Winding direction	Correct / Reversed (check) / Not judged
Axis deviation (relative to the head orientation)	Deviation from the typical direction (°)

Columns added at the end of lumen.csv (columns 31–34, cochlea rows only)

Column	Contents
winding_deg	Winding angle of the centerline (degrees)
axis_az_deg, axis_el_deg, axis_off45_deg	Azimuth and elevation of the axis relative to the head orientation, and the deviation from the typical direction (degrees)

When a warning appears

For an ear with the winding-direction warning, do not use the scala vestibuli / scala tympani side values until you correct the apex position and contamination and analyze again. With the axis-direction warning, check the seed and apex positions and contamination. Correct contamination with “Remove overlap in two planes”. If you open a ZIP saved by an earlier version and save it without reanalysis, these columns are empty.

LUMEN: Compare on a common shape (normalization on the skeleton)

What normalization on the skeleton is (1.3.0)

The skeleton of 1.2.4 (page 29) was used only for image orientation and warnings. In 1.3.0, the skeleton fitted to each ear serves as a ruler to map the cochlear fluid onto a common shape (normalization), so ears of different shapes can be compared at the same angles and positions. Each cochlear ROI pixel gets the angle of the nearest skeleton point (angle along the skeleton); within 1.5 mm of the skeleton is “in the tube”, and farther fluid is “Fluid outside the shape”. The skeleton, the fitting, existing values (volume ratios, LUMEN summary indices, E_SV / E_ST, θ, stretched view, LUMEN-V) and existing warning conditions do not change.

Origin of the angle and the turns

The angle along the skeleton is 0° at the basal end (including the hook region); its origin differs from the LUMEN-C angle (origin θ0, page 18). Turns are divided by this angle: basal 0–360°, middle 360–720°, apical from 720°. Left ears use the mirrored skeleton, with values and angles defined the same on both sides. Ears where the skeleton could not be fitted are not computed (table: “Fit failed”; section: “The skeleton could not be fitted”).

Where the section is, and tabs

Each LUMEN result frame has the section “Compare on a common shape” below LUMEN-C and LUMEN-V, with three figures side by side. If it is narrower than 540 px, the tabs “Untangled spiral”, “Common stretched view” and “Per-angle profile” show one at a time. Left ears are mirrored to match right ears.

The three figures

Figure	Contents
Untangled spiral	A schematic mapping the angle along the skeleton onto a spiral with constant spacing between turns (0° at the right, counterclockwise). The band is colored every 30° by the endolymph fraction in the tube; gray is “Too few samples”. Magenta dots are “Fluid outside the shape (not in the fraction)”, placed outside or inside the band at the nearest angle. The figure shows the overall value, the fluid outside the shape and the reliability
Common stretched view	The image read along the skeleton. Horizontally the angle along the skeleton, the same width for every ear; vertically the cochlear-axis direction (“Top: scala vestibuli side”, ± the tube thickness). Colors as in the LUMEN-C overlay row (red: endolymph, cyan: perilymph, black: outside the fluid). The magenta bars below are “Fluid outside the shape (per 10°)”
Per-angle profile	Horizontally the angle along the skeleton; vertically the endolymph fraction in the tube per 30° (%). The background is shaded by turn. If the other ear has a current result, its line is overlaid thinly (“This ear: whole …”, “Other ear: whole …”)

Cautions when reading the figures

Clinical images have too few pixels for the per-30° fractions in the figures, so these are obtained by resampling the image on discs crossing the tube at each skeleton point. The overall values in the figures and in the table are pixel counts. The untangled spiral is a schematic, not the actual shape of the cochlea. For ears with low reliability, every figure shows “Reliability: Low” in a red frame. The saved ZIP holds only the values, not the data for drawing the figures, so after opening saved results, reanalyze with “Start analysis” to show the figures (page 34).

LUMEN: Compare on a common shape: values and reliability

Rows added to the results table (cochlea)

Row name	Meaning
Endolymph fraction in the tube	Proportion (%) of pixels with negative HYDROPS in the cochlear fluid inside the tube (pixels of the cochlear ROI; as for the volume ratio, no MRC threshold or IAC-side exclusion). Fluid outside the shape is in neither the numerator nor the denominator. With input types B and C, the same judgment on that type's image
Basal turn, Middle turn, Apical turn (reference)	The same value per turn. The apical turn has few pixels and fluctuates, so it is for reference
Fluid outside the shape	Proportion (%) of the cochlear fluid outside the tube. Fluid on the IAC side is also counted here
Normalization reliability	High, Medium or Low, sometimes with a size note (e.g., High (size 0.80×, at the lower limit of 0.8–1.2)). Ears where the fit failed show “Fit failed”, ears where the calculation itself failed “Not computed”, and records from before 1.3.0 “—” in all six rows

How normalization reliability is decided (provisional)

Level	Condition
Low	The proportion of skeleton points inside the fluid (skeleton_fit) is below 0.95, the proportion of the cochlear fluid inside the tube is below 75%, or there is a departure or up/down reversed warning (page 30)
Medium	Not Low, and the proportion inside the tube is below 80%
High	Otherwise
Size note	When the skeleton scale is at the lower limit 0.80× or the upper limit 1.20×, a note is added without lowering the reliability

Ears with low reliability

Below the result frame, the strong warning “The reliability of the normalization on the skeleton is low, so do not use the “endolymph fraction in the tube” of this ear. Check the center points, the apex position, and contamination by the internal auditory canal or CSF.” appears with the criteria that failed (e.g., “(Reason: points inside the fluid 0.949, fluid inside the tube 74.9%)”; values are truncated so they never appear rounded to the threshold). skeleton_norm_low_confidence is added to warnings in lumen.csv and report.json. The values are still displayed and recorded.

About the criteria

The tube thickness of 1.5 mm and the reliability criteria are provisional, and their validity has not been verified; they will be adjusted. Whether the endolymph fraction in the tube reflects the degree of hydrops, normal ranges, and normalization on other scanners and protocols have not been confirmed.

LUMEN: Compare on a common shape: saved records

Added to the saved ZIP

File	Contents
lumen.csv (columns 39–45, cochlea rows only)	skeleton_norm_el_pct, _basal_el_pct, _middle_el_pct, _apical_el_pct (endolymph fraction in the tube, overall and per turn), skeleton_norm_out_pct (fluid outside the shape), skeleton_norm_confidence (high / mid / low), skeleton_norm_tube_mm (tube radius, mm)
lumen_skeleton_bins.csv	One row per ear and per 10° of angle along the skeleton: pixel counts of fluid inside the tube, of endolymph among it and of fluid outside the shape (vox_in, vox_el, vox_out), and the counts resampled on discs (smp_in, smp_el). Being counts, they can be summed over other intervals
report.json	skeleton_norm in each ear's LUMEN record (values, tube thickness, reliability with reasons and notes, scale, the criteria used (rule), counts per 10°)
LUMEN_NORM_SPIRAL / _STRETCH / _PROFILE_R, _L.png	Images of the three figures. Same rules as the 1.2.5 LUMEN images (page 21); only for ears that could be normalized. The heading summary is the endolymph fraction in the tube and the reliability. The per-angle profile includes the other ear's line
README.txt	Descriptions of the new columns, file and images, and lines for ears without images (ears that get them when reanalyzed, ears where the skeleton could not be fitted, ears where normalization could not be computed, ears where the LUMEN-C analysis failed)

ZIPs saved by earlier versions

If you open a ZIP saved by a version before 1.3.0 and do not reanalyze, the six table rows show “—”, the seven lumen.csv columns are empty, and lumen_skeleton_bins.csv and the images are not included. Reanalyze with “Start analysis” before saving to include them. 1.2.4 and 1.2.5 can also open 1.3.0 ZIPs with “Open saved results” (the new columns, lumen_skeleton_bins.csv and the new images are not loaded).

Reopening a ZIP saved by 1.3.0

As up to 1.2.5, ears with ① and ② done show “The saved results include LUMEN results. Press “Start analysis” to show them (the numbers will be the same).” (page 16). Reanalyzing with “Start analysis” also redraws the figures. If you save again without reanalysis, lumen_skeleton_bins.csv is rebuilt from report.json, and the images are carried over by the same rules as in 1.2.5.

ZIPs saved by the other app

LUMEN images in ZIPs saved by the other app (HYDROPS Studio) are not carried over when saved again without reanalysis, and README.txt has a line for those ears. Reanalyze with “Start analysis” before saving to include the images drawn by ELH Studio.